Auto Ticket Classification using FLOW DESIGNER

Phase 2: Requirement Analysis Phase

1. Introduction

Requirement Analysis is the second phase of the Auto Ticket Classification using FLOW DESIGNER project. This phase converts the initial ideas identified during brainstorming into clear, measurable, and implementable requirements. The purpose is to understand what the system must do, who will use it, what information it needs, and what conditions it must satisfy.

2. Purpose of the System

The proposed system will automatically analyze incoming support tickets and classify them into predefined categories using a FLOW DESIGNER workflow. After classification, the ticket can be routed to the appropriate team or queue. Tickets that cannot be confidently classified should be directed to a manual-review process.

3. Stakeholder Requirements

Stakeholder	Requirement / Expectation
Ticket Submitter	Ticket should be accepted with the necessary information and receive appropriate ack
Support Agent	Tickets should arrive in the correct category or queue with relevant information visib
Team Lead	The workflow should support consistent routing and provide useful status informatio
Administrator	Classification categories and routing rules should be maintainable.
Process Owner	The workflow should support monitoring and future improvement.

Figure 1. Stakeholders and their high-level requirements

4. Functional Requirements

ID	Functional Requirement
FR-01	The system shall receive a new support ticket as input.
FR-02	The system shall read relevant ticket fields such as subject, description, priority, and other available attri
FR-03	The workflow shall validate required ticket information before classification.
FR-04	The workflow shall apply predefined classification rules or decision logic.
FR-05	The system shall assign the ticket to an appropriate category when a valid match is found.

FR-06	The system shall route the classified ticket to the appropriate team or queue.
FR-07	The system shall identify tickets that cannot be classified automatically.
ID	**Functional Requirement**
FR-08	Unclassified or ambiguous tickets shall be routed for manual review.
FR-09	The workflow shall record the classification and routing outcome.
FR-10	The workflow shall support error handling when required data or processing steps fail.

Figure 2. Functional requirements of the proposed system

5. Non-Functional Requirements

Category	Requirement
Performance	The workflow should process tickets within an acceptable time for normal operational
Reliability	The workflow should handle failures without losing important ticket information.
Usability	The flow should be clear and understandable for administrators and support teams.
Maintainability	Classification and routing rules should be easy to review and update.
Scalability	The design should allow additional ticket categories and rules to be added.
Security	Ticket information should be handled according to organizational access and data-prote
Auditability	Important classification and routing actions should be traceable.

Figure 3. Non-functional requirements

6. Input Requirements

• Ticket subject or title.

• Ticket description or issue details.

• Priority or urgency, when available.

• Requester or source information, where required.

• Existing category, service, product, or issue-related fields, when available.

7. Output Requirements

• Assigned ticket category.

• Routing destination such as team or queue.

• Classification status such as Classified or Manual Review.

• Processing or classification result for tracking.

• Error or exception information when processing cannot be completed.

8. Classification Rules – Initial Requirement

The first version of the workflow can use rule-based classification. Keywords and ticket attributes may be mapped to predefined categories. For example, tickets containing terms related to password access can be assigned to an account or access category, while tickets describing a system outage can be assigned to a technical-support category. The actual categories and keywords should be finalized with the project stakeholders before implementation.

9. Exception and Manual Review Requirements

• If mandatory ticket information is missing, the workflow should flag the ticket for correction ormanual handling.

• If multiple categories match with no clear decision, the ticket should be sent for manual review.

• If no classification rule matches the ticket, it should be routed to a predefined fallback queue.

• Workflow errors should be recorded so that administrators can investigate and correct them.

10. Assumptions and Constraints

• Ticket data is available to the workflow through the selected ticketing system or integration.

• Initial categories and routing destinations are defined by the project team or process owner.

• Users have appropriate permissions to submit, view, classify, and route tickets.

• The accuracy of rule-based classification depends on the quality and completeness of ticketinformation.

• The first release may require periodic review and refinement of classification rules.

11. Requirement Traceability – Initial View

Requirement Area	FLOW DESIGNER Activity	Expected Result
Ticket Intake	Trigger / input handling	New ticket enters the workflow.
Validation	Condition / decision	Required information is checked.
Classification	Rules / decision logic	Category is identified.
Routing	Assignment / action	Ticket reaches the correct queue.
Exception	Fallback branch	Ambiguous tickets go to manual review.
Audit	Record / update action	Result can be tracked.

Figure 4. Initial requirement-to-workflow traceability

12. Acceptance Criteria

• A valid ticket can enter the workflow successfully.

• Required ticket information is checked before classification.

• A ticket matching a defined rule receives the expected category.

• A classified ticket is routed to the configured destination.

• An unmatched or ambiguous ticket follows the manual-review path.

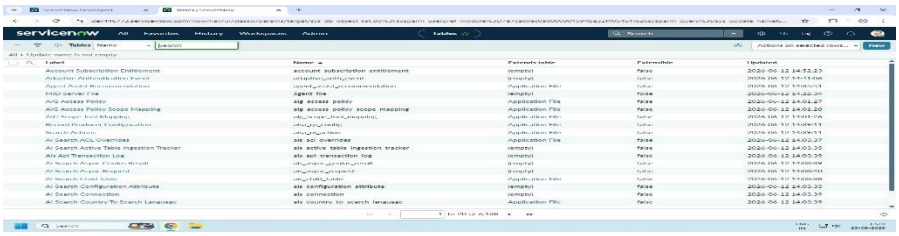
• The workflow records the classification and routing outcome.

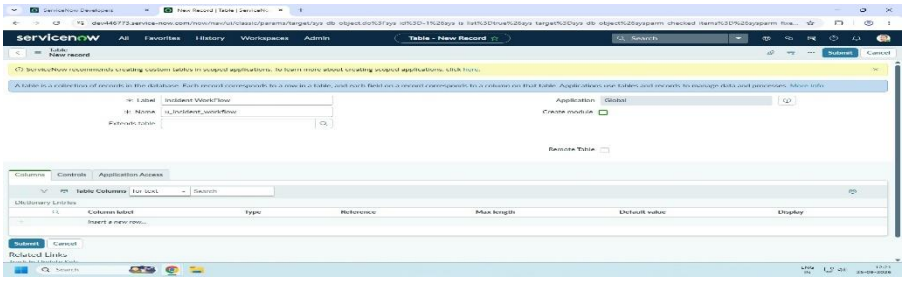
• Errors are handled without silently losing the ticket.

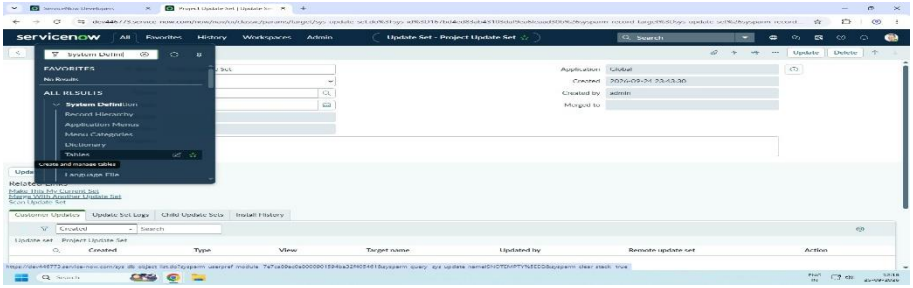
13. Phase 2 Outcome

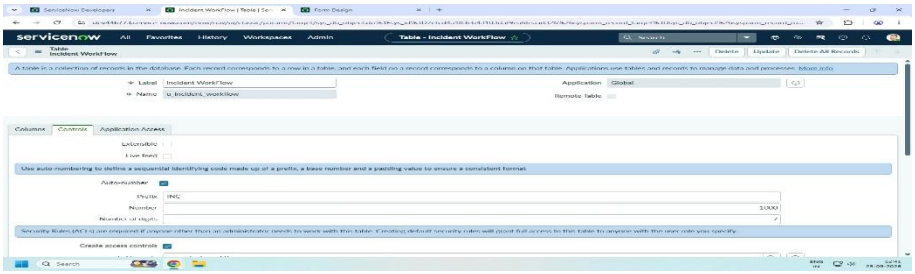
The Requirement Analysis Phase establishes the functional and non-functional requirements for Auto Ticket Classification using FLOW DESIGNER. It defines the expected inputs, outputs, classification behavior, routing process, exception handling, stakeholder needs, and acceptance criteria. These requirements provide a structured baseline for the next phase, where the detailed workflow design and implementation approach can be developed.

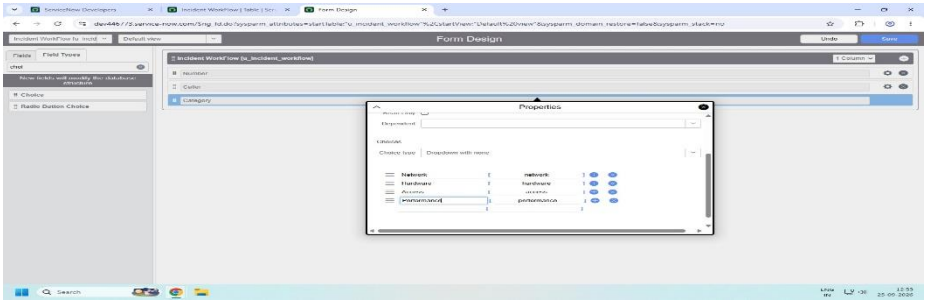
14.Screenshots

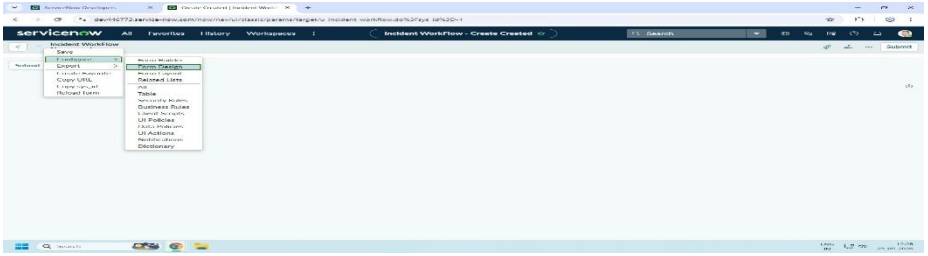


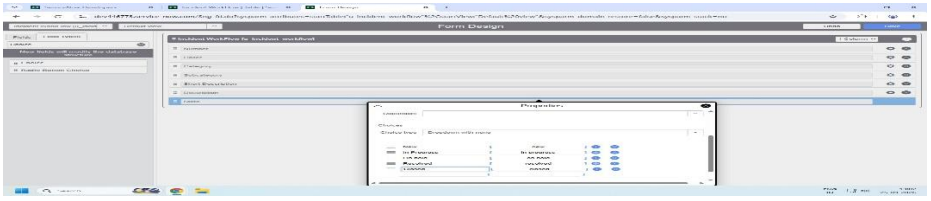


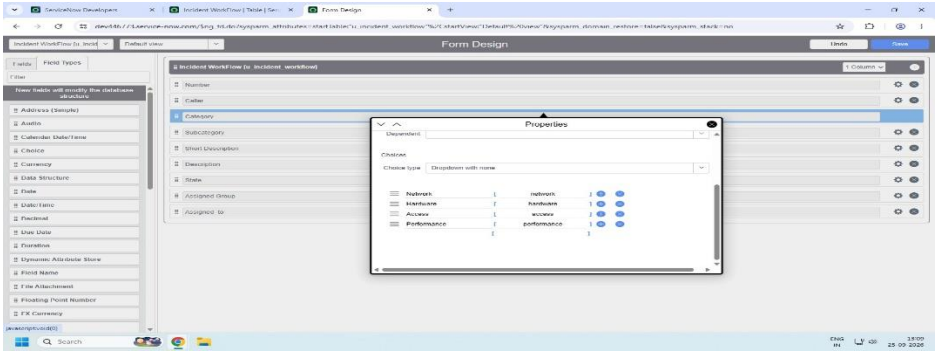


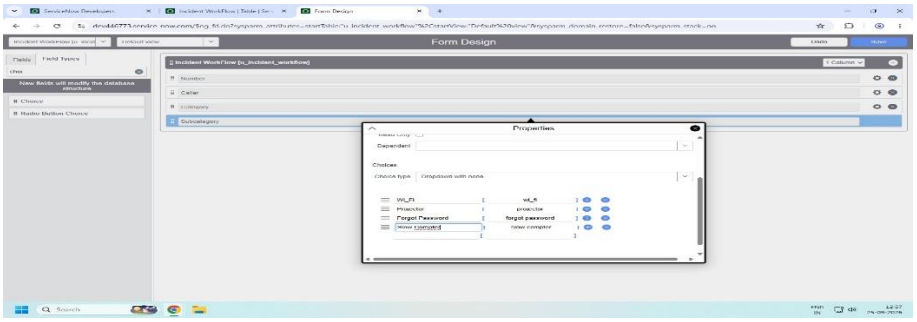


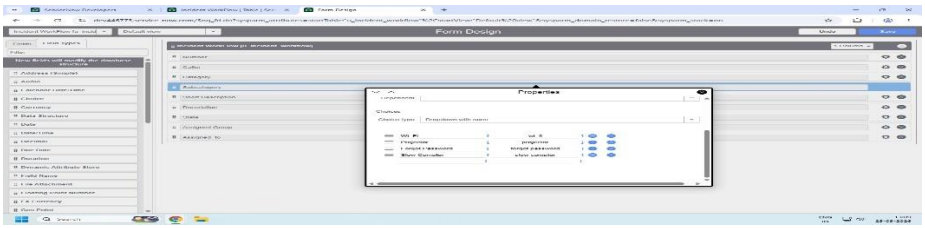












15.Conclusion

A clear requirement specification is essential for building a reliable automated ticket-classification workflow. By defining what the system should do before implementation, the project team can reduce ambiguity, align stakeholder expectations, and create a workflow that is easier to test and maintain.